Mihretab Nigatu Afework

Software Engineer | Python | Backend

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EXPERIENCE

Lead Back-end Engineer, Wiz Aroma **March 2025 - Present, Remote**

- Architected a distributed **7-bot Telegram platform** automating ordering, dispatch, and operations, improving order completion **38%** while processing **10k+ daily requests and 180+ peak orders/day**.
- Built a **state-driven Firebase backend** with SLA dispatch automation (300s timeout + rebroadcast) and optimized DB access patterns, reducing **API latency 57%** (420→180 ms) while maintaining **99.99% uptime**.

Technologies: Python, Firebase, Telegram Bot API, Docker, Railway, Git, Linux

Ful-Stack Developer, ATMA Oxygen **August 2023 - February 2024, Hawassa**

- Engineered **REST production data pipelines** managing **50k+ manufacturing records**, reducing data-entry errors **45%** via schema validation.
- Implemented **real-time monitoring and optimized DB indexing**, improving anomaly detection 30%, cutting downtime to **~5 hrs/week**, and accelerating historical queries ~60%.

Technologies: Python, SQL, REST APIs, relational databases, schema validation, analytics dashboards

Software & Database Systems Developer, Turbo Driving Center **May 2023 - July 2023, Hawassa**

- Built a Node.js + **MySQL scheduling platform** replacing manual booking, reducing scheduling errors **60%** and saving **~12 admin hours/week**.
- Designed a centralized trainee database managing **800+ active users and 15k+ records** with automated backups securing **20GB+ data** and **99.99% availability**.

Technologies: Node.js, JavaScript, MySQL, SQL, backup/DR automation, database integrity controls

PROJECTS

Social Vision Facial Recognition Search Engine (GitHub)

- Built **dual-embedding facial recognition pipelines** (DeepFace + dlib) with scalable indexing, image/video/live ingestion workflows, and **FastAPI search services**.
- Designed **multi-layer storage architecture** (Firebase + Firestore + local cache) with Docker optimization (~**60% faster builds**) and search latency **<6s** for <1k faces.

Technologies: Python, FastAPI, Streamlit, DeepFace, TensorFlow, OpenCV, Firestore, Firebase Docker

Miehab Voice Assistant (GitHub)

- Developed an **AI-powered** Python **voice assistant** integrating speech recognition, **LLM-driven NLP**, and external APIs to deliver conversational responses and actionable automation.
- Built a real-time, modular command system with **context-aware follow-ups**, plus web and voice interfaces using dual **TTS with STT fallback** for more reliable, natural integration across environments.

Technologies: Python, FastAPI, Speech_recognition, Transformers/LLM APIs, Wikipedia API, OpenWeather API, edge-tts

Personal Library Management System (GitHub)

- Designed a Django **relational system** managing books, borrowers, and lending workflows with search/filter pipelines and reporting dashboards.
- Implemented validation layers and **69 automated** tests, ensuring reliable transactions and inventory consistency.

Technologies: Django, SQLite, Bootstrap, Docker

SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS

Web and Backend: React, Bootstrap, Streamlit, Node.js, Express.js, Django, FastAPI, REST APIs, Telegram Bot API

Database: Firebase, Firestore, PostgreSQL, MySQL, MongoDB, SQL Server, SQLite, Data Modeling, Query Optimization

AI/ML & CV: Face Detection Pipelines, Facial Embeddings, Deep Learning Models, DeepFace, TensorFlow, Keras, PyTorch, OpenCV, dlib/face_recognition, Real-ESRGAN

DevOps: Docker, Docker Compose, Linux, Git, CI/CD, Logging Systems, Performance Monitoring

EDUCATION

BSc in Computer Science

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